Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password=P@ssw0rd
secret/my-secret created

C:\Users\NK>kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-04-28T07:19:14Z"
  name: my-secret
  namespace: default
  resourceVersion: "4106"
  uid: 7cb5e7a6-b65d-4b59-8048-2ef60df9f115
type: Opaque

C:\Users\NK>echo|set /p="my-username"> username.txt

C:\Users\NK>echo|set /p="my-password"> password.txt

C:\Users\NK>kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
secret/my-file-secret created

C:\Users\NK>kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-04-28T07:19:32Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "4122"
  uid: 2f953765-d6fa-4ef0-bf19-f339d45b6daf
type: Opaque
```

```
C:\Users\NK>kubectl exec pod-with-secret -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=pod-with-secret
USERNAME=admin
PASSWORD=P@ssw0rd
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
NGINX_VERSION=1.29.8
NJS_VERSION=0.9.6
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root

C:\Users\NK>kubectl delete secret my-secret
secret "my-secret" deleted from default namespace

C:\Users\NK>kubectl delete secret my-file-secret
secret "my-file-secret" deleted from default namespace

C:\Users\NK>kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted from default namespace

C:\Users\NK>kubectl delete secret my-manifest-secret
Error from server (NotFound): secrets "my-manifest-secret" not found

C:\Users\NK>del username.txt password.txt

C:\Users\NK>kubectl get secrets
No resources found in default namespace.
```